

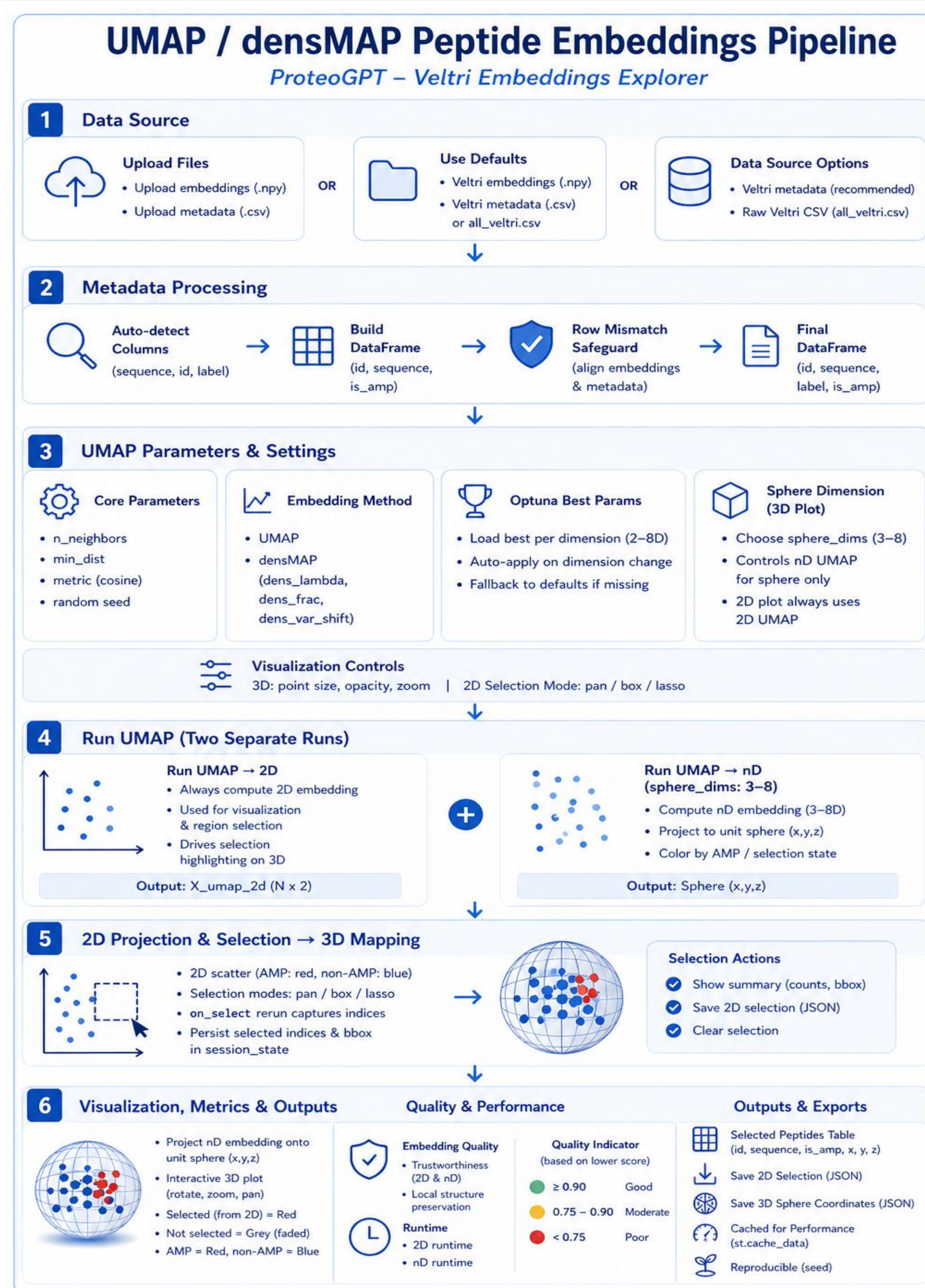
Abstract

Antimicrobial resistance (AMR) is a growing global health threat, increasing the need for new therapeutics such as antimicrobial peptides (AMPs). However, AMP discovery remains difficult because peptide sequence space is large, complex, and costly to screen experimentally. This work proposes a visual analytics framework that uses peptide-language model embeddings to represent peptide sequences and facilitate the exploration of their similarities. By projecting these embeddings into a lower-dimensional space, the framework enables interpretable cluster analysis and identifies promising regions for AMP design.

Introduction

- There is an urgent need for new antimicrobial agents, particularly alternatives such as AMPs.
- The discovery of AMPs is challenging due to the large, discrete nature of peptide sequence space and the high experimental costs associated with testing.
- Peptide language models (PLMs) are capable of representing peptide sequences as fixed-size embeddings that capture both functional and semantic similarities.
- Visual analytics enhances interpretability by projecting high-dimensional embeddings into two- or three-dimensional spaces.

Methodology / Approach



Results

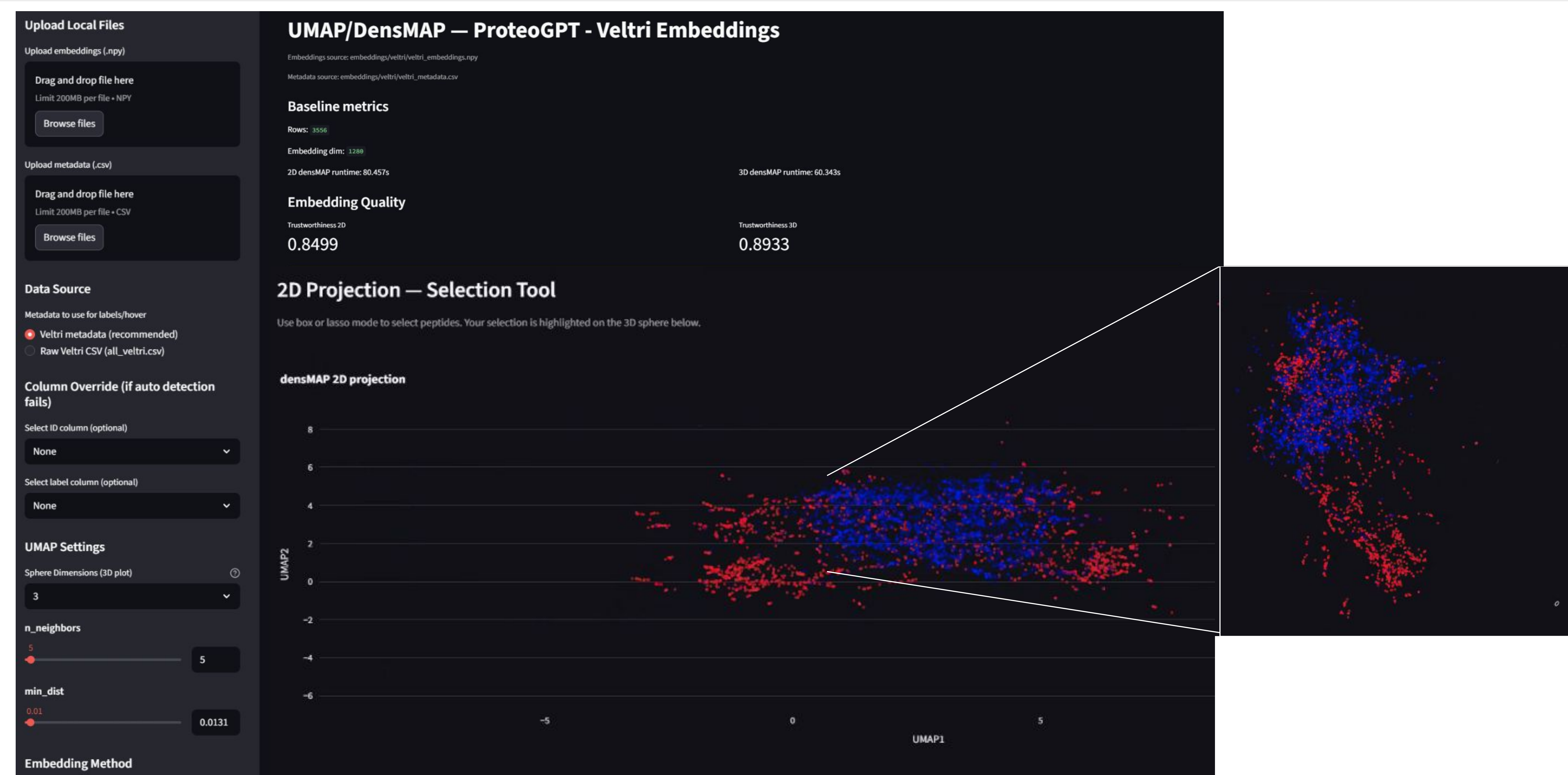


Figure 1. 2D and 3D UMAP/densMAP projections of ProteoGPT peptide embeddings. The left panel shows the 2D embedding used for region selection, while the right panel displays the corresponding high-dimensional structure projected onto a 3D unit sphere. Selected regions in 2D are mapped directly onto the sphere, revealing underlying spatial relationships and clustering patterns between antimicrobial peptides (red) and non-AMPs (blue). This dual-view approach enables interactive exploration while preserving local structure (trustworthiness ≈ 0.85).

Conclusion

PepSpace, a visual analytics framework for exploring peptide embeddings. By combining peptide language models with dimensionality reduction, our approach improves interpretability of peptide space and supports the identification of promising antimicrobial candidates.

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